

Snoring

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Snoring is a raspy noise produced in the nose and throat during sleep. It is quite common and becomes more common as people age. Very many men and many women snore. However, what qualifies as snoring depends on the person listening to it, and how loudly and how much a person snores vary from night to night. Thus, the percentage of people who snore is only an estimate.

(See also [Overview of Sleep](#).)

A few people snore lightly, but snoring is usually noticeable and is sometimes loud enough to be heard in another room. Snoring is distressing usually only to other people, typically a bed partner or roommate trying to sleep. Snorers seldom know that they snore unless others tell them. However, some hear their own snoring as they wake up.

Snoring can have significant social consequences. It frequently causes stress between the snorer and bed partner or roommates.

Other symptoms such as waking up frequently, gasping or choking during sleep, [excessive daytime sleepiness](#), and morning headache may also be present, depending on what is causing the snoring.

Snoring results from fluttering of soft tissues in the throat, particularly the soft palate (the back part of the roof of the mouth). The fact that people do not snore when they are awake suggests that relaxation of muscles during sleep is part of the cause. This relaxation is thought to decrease the stiffness of tissue, making it more likely to flutter (just as a cloth flag is more likely to flutter in a breeze than is a similar-sized sheet of metal). Also, tissue relaxation narrows the upper airway, making flutter more likely.

Causes of Snoring

Primary snoring is snoring that does not cause people to wake up more often than normal during the night. During sleep, the amount of airflow into the lungs and oxygen level in the blood are normal. Because these factors are normal, people do not have excessive daytime sleepiness.

Sleep-disordered breathing

Snoring is often a symptom of sleep-disordered breathing. Sleep-disordered breathing ranges from upper airway resistance syndrome to **obstructive sleep apnea** (OSA). These conditions differ mainly in how much of the airway is blocked (degree of airway obstruction) and where the blockage is. The effects involve mainly disturbances of sleep and/or airflow.

Every hour during sleep, people with OSA have 5 or more brief episodes when they stop breathing or when their breathing is very shallow. They also have one or more of the following:

- Daytime sleepiness, episodes of unintentionally falling asleep, unrefreshing sleep, fatigue, or insomnia
- Waking up with breath holding, gasping, or choking
- Reports by a bed partner of loud snoring, interruptions in breathing, or both during the person's sleep

Upper airway resistance syndrome causes excessive daytime sleepiness or other symptoms but not all that is required for doctors to diagnose OSA. For example, it, unlike OSA, does not cause breathing to stop or become shallow many times during sleep.

Complications

Whether snoring itself has adverse effects is unclear. However, people who have OSA have an increased risk of high blood pressure, stroke, heart disorders, and diabetes.

Risk factors

Risk factors for snoring include

- Older age (over 50)
- Obesity, particularly fat distributed around the neck or midriff

- Use of alcohol (a very common cause of snoring) or other sedatives
- Long-term (chronic) nasal congestion
- A small jaw or a jaw that is farther back than normal
- Menopause
- Male sex
- Pregnancy
- Abnormalities that can block airflow, such as large tonsils, a large tongue, a large soft palate, a deviated nasal septum, and nasal polyps

Snoring frequently occurs in families.

Evaluation of Snoring

For doctors, the main goal of evaluation is identifying people who are at high risk of obstructive sleep apnea (OSA). Not all people who snore have OSA. However, most people with OSA do snore.

Warning signs

The following symptoms are cause for concern:

- Episodes of not breathing or of choking during sleep (as witnessed by a bed partner)
- Headaches when waking up in the morning
- Sleepiness during the day
- [Obesity](#)
- Very loud, constant snoring
- [High blood pressure](#)

These symptoms may suggest sleep apnea.

When to see a doctor

People with warning signs should see a doctor soon because testing may be needed.

People without warning signs are unlikely to need testing and may wish to try [general measures](#) to help reduce snoring before they call a doctor. If these measures are unsuccessful and snoring is very bothersome to their bed partner, people should see their doctor.

What the doctor does

Doctors first ask questions about the snoring and other symptoms and then about the person's other medical history. Because several important findings are noticed mainly by others, doctors try to interview the bed partner or roommates when possible.

Doctors then do a physical examination. What they find during the history and physical examination helps them decide whether tests for OSA need to be done.

Doctors ask how severe the snoring is. For example, they may ask the bed partner

- Whether the person snores every night and, if not, how many nights
- Whether the person snores all night long and, if not, how much of the night
- How loud the snoring is

The person and bed partner are also asked to describe

- How often the person seems to wake up during the night
- Whether the person has stopped breathing or had episodes of gasping or choking
- Whether sleep seems unrefreshing or the person has morning headaches
- How sleepy the person is during the day

Doctors also ask about disorders that may be associated with [obstructive sleep apnea](#) (OSA), particularly [high blood pressure](#), heart disorders, [stroke](#), [acid reflux](#), [atrial fibrillation](#) (an abnormal heart rhythm), and [depression](#). They ask about the amount of alcohol the person drinks, including how close to bedtime the person drinks it. Whether the person takes any sedating or muscle-relaxing medications is also important.

During the physical examination, doctors measure the person's height and weight in order to calculate [body mass index](#) (BMI). The higher the person's BMI, the greater the risk of OSA. Doctors may measure neck size. OSA is more likely when the neck is larger than about 16 inches for women and about 17 inches for men.

Doctors also inspect the nose and mouth for signs of airway obstruction and risk factors for snoring—for example, nasal polyps, a deviated septum, chronic nasal congestion, a high and arched palate, a jaw that is small or farther back than normal, and an enlarged tongue, tonsils, or uvula (the structure that hangs down at the back of the throat). Doctors measure blood pressure because OSA is more likely when blood pressure is high.

Although doctors are not able to predict risk precisely, the more risk factors and warning signs people have, the greater their risk of OSA.

Testing

When doctors suspect obstructive sleep apnea (OSA), they typically do tests to confirm the diagnosis.

Testing consists of [polysomnography](#). For this test, people sleep overnight in a sleep laboratory while their breathing and other functions are monitored. A sleep laboratory may be located in a hospital, clinic, hotel room, or other facility that is equipped with a bed, bathroom, and monitoring equipment. Polysomnography can be done in the home (home sleep study) if people who snore do not have many other coexisting disorders. However, because snoring is so common and polysomnography is costly and time-consuming, doctors may not recommend polysomnography for everyone who snores. Testing people who have warning signs (particularly those with episodes of not breathing witnessed by another person) and those with several risk factors is particularly important.

If people without warning signs do not appear to have any sleep disturbance other than snoring, they typically do not need tests. However, they should schedule regular follow-ups so that their doctor can check for the development of such problems.

Treatment of Snoring

Causes of snoring, such as chronic nasal obstruction and [obstructive sleep apnea](#) (OSA), are treated.

For snoring itself, treatment includes general measures to eliminate risk factors and physical methods to open the upper airway and/or stiffen the involved structures.

General measures

Several general measures may help reduce primary snoring. None are effective in everyone, but some people may benefit. Measures include

- Sleeping with the head elevated
- Lying on one side (rather than the back)
- Losing weight
- Avoiding alcohol and sedating medications for several hours before bedtime

- Treating nasal congestion—for example, with a corticosteroid nasal spray or sometimes other medications

The best way to elevate the head is to put blocks under 2 of the bed's legs to raise the head of the bed or to use a wedge pillow that slants the whole upper body. People should not use pillows to raise only the head.

Measures that force the sleeper onto his/her side during the night could include, for example, attaching a tennis ball to the back of a person's night shirt.

Bed partners may benefit from using earplugs or white-noise machines. Sometimes alternate sleeping arrangements (such as a separate room) are necessary.

Oral appliances

Oral appliances are worn only during sleep. They include

- Mandibular advancement splints
- Tongue-retaining devices

These appliances, which must be fitted by specially trained dentists, can help keep the airway open during sleep in people with obstructive sleep apnea (OSA) and may help reduce snoring.

Mandibular advancement splints are small plastic devices, made by specially qualified dentists, that fit in the mouth like a mouth guard or orthodontic retainer. They pull the lower jaw (mandible) and tongue forward and thus help keep the airway open during sleep. Many of these devices can be adjusted in small increments to ensure the best results. Adjustable devices are more effective than those that cannot be adjusted, such as the "boil-and-bite" self-fitted oral appliances that can be bought over-the-counter.

Tongue-retaining devices use suction to keep the tongue forward. If the tongue moves back in the mouth, it can block the airway. These devices are more uncomfortable than mandibular advancement devices.

Oral appliances may be used alone or with other treatments for sleep-related breathing disorders, such as weight management, surgery, or continuous positive airway pressure.

Oral appliances can cause discomfort and excess salivation, and teeth may be pushed out of alignment. But most people tolerate them well.

Continuous positive airway pressure (CPAP)

With CPAP, people breathe through a small mask applied to the nose or to the nose and mouth. The mask is attached to a device that supplies air at a pressure that helps prevent the airway from narrowing or collapsing when people breathe in (which is when most snoring occurs).

CPAP provides very effective relief of obstructive sleep apnea (OSA) and helps reduce snoring, but it is rarely used to treat snoring without OSA. Some people find CPAP devices uncomfortable or inconvenient, but most people with OSA are comfortable using these devices. Close follow-up by a health care practitioner is needed during the first 2 weeks of use to make sure the CPAP mask fits correctly and to encourage people as they learn to sleep with the mask. People with OSA tend to be more motivated to use CPAP than those whose only problem is snoring because untreated OSA can cause significant symptoms and increase the risk of heart disorders and stroke.

Surgery

Some upper airway obstructions that contribute to snoring, such as nasal polyps, enlarged tonsils, and a deviated septum, can be treated surgically. But whether and how well such procedures reduce snoring have not been proved.

In addition, a number of surgical procedures have been developed specifically to treat OSA, and some can help reduce snoring. These procedures reshape tissues of the palate and/or uvula or stiffen the palate using implants or injections. They include uvulopalatopharyngoplasty, laser-assisted uvuloplasty, injection snoreplasty, radiofrequency ablation, and palatal implants. More studies are needed to determine whether these surgical procedures effectively treat snoring in the long term.

In **uvulopalatopharyngoplasty**, tissues of the palate and uvula are surgically reshaped. Excess tissue is removed, and the airway is widened. This procedure requires a general anesthetic and a stay in the hospital. It can reduce snoring, but its effects may last only a few years.

The following procedures do not require a hospital stay and require only a local anesthetic.

For **laser-assisted uvuloplasty**, a laser or high-energy microwave device is used to reshape tissue, so this procedure is less invasive than uvulopalatopharyngoplasty. However, whether it can reduce snoring has not been proved, although some people seem to benefit from it.

For **injection snoreplasty** (a form of sclerotherapy), a substance that irritates the tissues and causes fibrous scar tissue to form is injected into the soft palate. As a result, the soft palate and uvula become stiffer and less likely to vibrate. Whether this procedure can reduce snoring requires further study.

For **radiofrequency ablation**, a probe is used to deliver heat (from an electrical current) into the soft palate. This procedure shrinks and stiffens tissues. It can reduce snoring, but further study is needed.

Palatal implants, made of polyethylene, can be surgically placed in the soft palate to stiffen it. Three small implants are used. Whether these implants are useful for snoring alone has not been proved.

A recently developed, removable **tongue muscle stimulator** is intended to help tongue muscles function better in people with snoring or OSA. The device is worn in the mouth and contains electrodes that stimulate nerves and muscles in the tongue. It can be used during the day. However, more studies are needed to determine how effective it is.

Key Points

- Not all people who snore have obstructive sleep apnea (OSA), but most people who have OSA snore.
- Warning signs, such as episodes of not breathing or of choking during sleep, daytime sleepiness, and obesity, help identify people at risk of OSA and thus in need of testing with polysomnography.
- General measures to manage snoring include avoiding alcohol and sedating medications before bedtime, sleeping with the head elevated, losing weight, and, for the bed partner, using earplugs and having alternative sleeping arrangements.
- Specific treatments for snoring include devices to keep the airway open (such as a mandibular advancement splint) and surgery.
- Doctors also treat causes of snoring, such as chronic nasal obstruction and OSA, which may be treated with continuous positive airway pressure (CPAP).



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